

Daily Quiz 10 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10

Name * 

test

Entry Number * 

test

✓ **Correct** 1/1 Points

ld r5, 12[r3]

Which sequence correctly describes its passage through the processor? 

- ☐ IF computes the effective address; OF accesses data memory; EX writes r5; MA computes pc + 4.
- ☐ OF reads r3 and r5; EX compares them; MA stores r5; RW writes the address to r3.

- ☐ OF reads r5 and r3; EX reads data memory; MA adds the offset; RW increments the PC.
- ☒ OF reads r3 and forms the extended immediate; EX computes the effective address; MA reads data memory; RW writes `ldResult` to r5.

✓ **Correct** 1/1 Points

Which statements about the processor are correct? 

Select all that apply.

- ☒ `isBranchTaken` is generated in EX and used in IF to select the next PC.
- ☐ The OF stage updates flags . E and flags . GT when it decodes a `cmp` instruction.
- ☒ The register file, memories, and ALU are datapath elements.
- ☒ The control path generates signals that govern the movement and processing of data.

✓ **Correct** 1/1 Points

Immediately before a branch decision:

`flags.E = 0`

`flags.GT = 1`

What is the tuple of `isBranchTaken` values for: (`beq`, `bgt`, `b`, non-branch) respectively? 

- ☐ (1, 1, 1, 0)
- ☒ (0, 1, 1, 0)
- ☐ (0, 1, 0, 1)
- ☐ (1, 0, 1, 0)

✓ **Correct** 1/1 Points

During a `ret` instruction, register r15 contains `0x1A40` while the embedded-target logic happens to produce `0x2200` What are (`branchPC`, next PC)? 

- ☒ (0x1A40, 0x1A40)
- ☐ (0x1A40, pc + 4)
- ☐ (0x2200, 0x1A40)
- ☐ (0x2200, 0x2200)

✓ **Correct** 1/1 Points

The processor executes `st r6, 12[r2]` with `r2 = 0x00001000`
`r6 = 0xDEADBEEF`. Which values reach `mar` and `mdr` in the MA stage? 

- ☒ `mar = 0x0000100C`, `mdr = 0xDEADBEEF`
- ☐ `mar = 0x00001000`, `mdr = 0xDEADBEEF`
- ☐ `mar = 0x0000100C`, `mdr = 0x0000000C`
- ☐ `mar = 0xDEADBEEF`, `mdr = 0x0000100C`

✓ **Correct** 1/1 Points

For an `add` instruction, EX receives:

`op1 = 9`

`op2 = 5`

`immx = -3`

If the I bit is 1, which ALU inputs and result are correct? 

- ☒ `A = 9`, `B = -3`, `aluResult = 6`
- ☐ `A = 9`, `B = 5`, `aluResult = 14`
- ☐ `A = 9`, `B = -3`, `aluResult = 12`
- ☐ `A = 5`, `B = -3`, `aluResult = 2`

✓ **Correct** 1/1 Points

For: st r8, 20[r3] and, separately, for: ret which tuple gives: (store read-port 1, store read-port 2, ret read-port 1)?



- ☐ (r3, r8, r14)
- ☐ (r8, r3, r15)
- ☒ (r3, r8, r15)
- ☐ (r3, r3, r15)

✓ **Correct** 1/1 Points

Which statements about the immediate and branch-target logic are correct?

Select all that apply.

- ☐ The 27-bit branch offset is added directly to the PC without any scaling.
- ☒ The branch offset is shifted left by two bits and sign-extended before being added to the PC.
- ☒ immx is obtained from the 18-bit immediate field and expanded to 32 bits.
- ☒ immx and branchTarget are computed irrespective of the instruction format.

✓ **Correct** 1/1 Points

For a call instruction, the RW unit receives:

pc = 0x00001000

aluResult = 0xAAAA0000

ldResult = 0xBBBB0000

rd = r4

With writeback enabled, which register write occurs?

- ☐ r4 <- 0xBBBB0000

- ☐ r15 <- 0xAAAA0000
- ☐ r4 <- 0x00001004
- ☒ r15 <- 0x00001004

✓ **Correct** 1/1 Points

An add instruction uses the adder, while transmission gates block new operands from unused ALU units. What is the intended behavior when the select input of such a gate is 0? 

- ☒ The unused unit is disconnected from the new input, avoids switching, and therefore dissipates less power.
- ☐ The unused unit still tracks every new input; only its final output is ignored.
- ☐ The unused unit is forced to zero and all of its internal state is erased.
- ☐ The new operands are broadcast to every unit to reduce the instruction latency.

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